Computational Finance Set 2

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1 Least-Squares Monte Carlo Simulation

Study the original paper that introduces least-squares Monte Carlo simulation for pricing American options (Longstaff and Schwartz, 2001). We use the 8 hard-coded paths from the paper with strike price $K = 1.10$ and risk-free rate $r = 6\%$.

Part (i) — Numerical Example with 8 Paths

5.8/6.0

1. **Cash-flow matrix at time 3, Regression at time 2, approximation formula for $E[Y \mid X]$ at time 2, and Optimal early exercise decision at time 2.**

   Cash-flow matrix at time 3

Cash-flow matrix at time 3

	t = 1	t = 2	t = 3
1	–	–	0.00000
2	–	–	0.00000
3	–	–	0.07000
4	–	–	0.18000
5	–	–	0.00000
6	–	–	0.20000
7	–	–	0.09000
8	–	–	0.00000

Figure 1: Cash-flow matrix at time 3.

Regression at time 2

Regression at time 2

Path	Y	X
1	0.00000	1.08000
2	–	–
3	0.06592	1.07000
4	0.16952	0.97000
5	–	–
6	0.18835	0.77000
7	0.08476	0.84000
8	–	–

Figure 2: Regression at time 2.

The least-squares approximation formula for $E[Y \mid X]$ at time 2 is obtained by regressing Y on a constant, X, and X^2 using only the in-the-money paths. The resulting conditional expectation function is:

$$E[Y \mid X] = a_0 + a_1 X + a_2 X^2$$

```
E[Y|X] = -1.06999 + 2.98341*X + -1.81358*X^2
```

Figure 3: Conditional expectation function at time 2.

Optimal early exercise decision at time 2

Optimal early exercise decision at time 2

Path	Exercise	Continuation
1	0.02000	0.03674
2	–	–
3	0.03000	0.04590
4	0.13000	0.11753
5	–	–
6	0.33000	0.15197
7	0.26000	0.15642
8	–	–

Figure 4: Optimal early exercise decision at time 2.

2. **Cash-flow matrix at time 2, Regression at time 1, approximation formula for $E[Y \mid X]$ at time 1, and Optimal early exercise decision at time 1.**

   Cash-flow matrix at time 2

   Cash−flow matrix at time 2

	t = 1	t = 2	t = 3
1	−	0.00000	0.00
2	−	0.00000	0.00
3	−	0.00000	0.07
4	−	0.13000	0.00
5	−	0.00000	0.00
6	−	0.33000	0.00
7	−	0.26000	0.00
8	−	0.00000	0.00

   Figure 5: Cash-flow matrix at time 2.

   Regression at time 1

   Regression at time 1

Path	Y	X
1	0.00000	1.09000
2	−	−
3	−	−
4	0.12243	0.93000
5	−	−
6	0.31078	0.76000
7	0.24486	0.92000
8	0.00000	0.88000

   Figure 6: Regression at time 1.

   The least-squares approximation formula for $E[Y \mid X]$ at time 1 is again obtained by regressing Y on a constant, X, and X^2 using only in-the-money paths. The resulting conditional expectation function is:

   $$E[Y \mid X] = a_0 + a_1 X + a_2 X^2$$

```
E[Y|X] = 2.03751 + -3.33544*X + 1.35646*X^2
```

Figure 7: Conditional expectation function at time 1.

Optimal early exercise decision at time 1

Optimal early exercise decision at time 1

Path	Exercise	Continuation
1	0.01000	0.01349
2	–	–
3	–	–
4	0.17000	0.10875
5	–	–
6	0.34000	0.28606
7	0.18000	0.11701
8	0.22000	0.15276

Figure 8: Optimal early exercise decision at time 1.

3. **Stopping rule and Option cash flow matrix.**

Stopping rule

Stopping rule

	t = 1	t = 2	t = 3
1	0	0	0
2	0	0	0
3	0	0	1
4	1	0	0
5	0	0	0
6	1	0	0
7	1	0	0
8	1	0	0

Figure 9: Stopping rule.

Option cash flow matrix

```
Option cash flow matrix
    t = 1   t = 2   t = 3
1 0.00000 0.00000 0.00000
2 0.00000 0.00000 0.00000
3 0.00000 0.00000 0.07000
4 0.17000 0.00000 0.00000
5 0.00000 0.00000 0.00000
6 0.34000 0.00000 0.00000
7 0.18000 0.00000 0.00000
8 0.22000 0.00000 0.00000
```

Figure 10: Option cash flow matrix.

4. **American put option price** (rounded to 5 decimal places).

```
The american put option price rounded to 5 decimals is:  0.11443
```

Figure 11: American put option price (rounded to 5 decimal places).

5. **European put option price** using the same 8 paths (rounded to 5 decimal places).

```
The european put option price rounded to 5 decimals is:  0.05638
```

Figure 12: European put option price (rounded to 5 decimal places).

Part (ii) — Replication of Table 1

4.0/6.0

1. **Replace hard-coded paths by Monte Carlo simulation.**

   We replace the 8 hard-coded paths by $N = 100{,}000$ (50,000 plus 50,000 antithetic) paths simulated under the risk-neutral GBM:

   $$dS = rS\,dt + \sigma S\,dZ,$$

   with strike price $K = 40$, and free-risk interest rate $r = 6\%$, and $M = 50$ exercise opportunities per year. We use a constant and the first three weighted Laguerre polynomials evaluated at the normalized stock price $x = S/K$ as basis functions:

$$L_0(x) = e^{-x/2}, \quad L_1(x) = e^{-x/2}(1-x), \quad L_2(x) = e^{-x/2}\left(1 - 2x + \frac{x^2}{2}\right).$$

The normalization $x = S/K$ is necessary to avoid numerical underflow in the exponential term when S ranges from 36 to 44, as noted in Section 8.3 of the paper.

2. **Replication of the columns in the red box of Figure 2** (Simulated American, s.e., Closed form European, Early exercise value), displaying 4 digits after the decimal point.

   The European put values are computed using the closed-form Black-Scholes formula:

$$P_{\text{BS}} = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1), \quad d_{1,2} = \frac{\ln(S_0/K) + \left(r \pm \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}.$$

   The standard error of the simulation estimate is $\widehat{\text{SE}} = s(M)/\sqrt{N}$, where $s(M)$ is the sample standard deviation of the discounted path-wise cash flows.

S	sigma	T	Simulated American	s.e.	Closed form European	Early exercise value
36	0.2	1	4.4714	(0.0094)	3.8443	0.6271
36	0.2	2	4.8229	(0.011)	3.7630	1.0599
36	0.4	1	7.1068	(0.0192)	6.7114	0.3954
36	0.4	2	8.5108	(0.0227)	7.7000	0.8108
38	0.2	1	3.2424	(0.0093)	2.8519	0.3905
38	0.2	2	3.7521	(0.011)	2.9906	0.7615
38	0.4	1	6.1412	(0.0186)	5.8343	0.3069
38	0.4	2	7.6694	(0.0222)	6.9788	0.6906
40	0.2	1	2.3117	(0.0087)	2.0664	0.2453
40	0.2	2	2.8921	(0.0106)	2.3559	0.5362
40	0.4	1	5.3138	(0.0181)	5.0596	0.2542
40	0.4	2	6.9163	(0.0218)	6.3260	0.5903
42	0.2	1	1.6248	(0.0077)	1.4645	0.1603
42	0.2	2	2.2042	(0.0097)	1.8414	0.3628
42	0.4	1	4.5816	(0.0173)	4.3787	0.2029
42	0.4	2	6.2363	(0.0213)	5.7356	0.5007
44	0.2	1	1.1131	(0.0065)	1.0169	0.0962
44	0.2	2	1.6861	(0.0087)	1.4292	0.2569
44	0.4	1	3.9323	(0.0162)	3.7828	0.1495
44	0.4	2	5.6110	(0.0206)	5.2020	0.4090

Figure 13: Replication of the columns in the red box of Figure 2.

2 Finite Difference Method

1.0/2.0

Part (i) — European Put via FTCS

Starting from the provided `ftcs_bs_put.py`, we adapt the code to price European puts for the same 20 parameter combinations as in Table 1 of Longstaff and Schwartz (2001). We use the discretisation settings of the paper: $N_x = 1{,}000$ stock-price steps and $N_t = 40{,}000 \times T$ time steps. Working in log-price space $m = \log S$, the Black–Scholes PDE becomes:

$$\frac{\partial P}{\partial t} = \frac{\sigma^2}{2}\frac{\partial^2 P}{\partial m^2} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial P}{\partial m} - rP.$$

The spatial grid is centred at $\log K$ to ensure adequate resolution near the strike:

$$m_j = \log K - \frac{L}{2} + jh, \quad h = \frac{L}{N_x}, \quad L = 2\log K + 2.$$

The FTCS (forward differences in time and central differences in space) evolution matrix is:

$$F = (1 - rk)\, I + \frac{\sigma^2 k}{2h^2} T_2 + \frac{k\left(r - \frac{\sigma^2}{2}\right)}{2h} T_1,$$

where T_2 and T_1 are the standard centred-difference matrices for the second and first spatial derivatives respectively, and $k = T/N_t$ is the time step.

The finite differences European prices compared to the Black-Scholes closed-form values are:

S	sigma	T	FD European	BS European	Difference
36	0.2	1	3.8441	3.8443	-0.0002
36	0.2	2	3.7628	3.7630	-0.0002
36	0.4	1	6.7112	6.7114	-0.0002
36	0.4	2	7.6999	7.7000	-0.0002
38	0.2	1	2.8518	2.8519	-0.0001
38	0.2	2	2.9904	2.9906	-0.0001
38	0.4	1	5.8342	5.8343	-0.0001
38	0.4	2	6.9787	6.9788	-0.0001
40	0.2	1	2.0656	2.0664	-0.0008
40	0.2	2	2.3553	2.3559	-0.0005
40	0.4	1	5.0592	5.0596	-0.0004
40	0.4	2	6.3257	6.3260	-0.0003
42	0.2	1	1.4641	1.4645	-0.0004
42	0.2	2	1.8411	1.8414	-0.0003
42	0.4	1	4.3785	4.3787	-0.0002
42	0.4	2	5.7355	5.7356	-0.0002
44	0.2	1	1.0166	1.0169	-0.0004
44	0.2	2	1.4289	1.4292	-0.0003
44	0.4	1	3.7826	3.7828	-0.0002
44	0.4	2	5.2018	5.2020	-0.0002

Figure 14: Finite differences European prices compared to the Black-Scholes closed-form values.

The finite differences European prices are very similar to those obtained with the Black-Scholes formula, confirming the correctness of the implementation.

Part (ii) — American Put via FTCS with Early Exercise

1.5/4.0

To price American puts we add a single line inside the time-stepping loop: after each FTCS step, the option value is compared with the intrinsic value $\max(K - S, 0)$ (and we decide if we exercise the option or we hold it):

```
for i in range(Nt):
    time2mat = i * k
    p = np.zeros(Nx-1)
    p[0] = ( 0.5*k*(sigma**2)/(h**2) - k*(r-0.5*(sigma**2))/(2*h) ) * K * np.exp(-r*time2mat)
    U[:, i+1] = np.dot(F, U[:, i]) + p
    U[:, i+1] = np.maximum(U[:, i+1], intrinsic)   #change, we exercise or we hold
```

Figure 15: Line that changes in the code.

This enforces the optimal stopping condition $V(S,t) \geq (K - S)^+$ at every grid node and every time step, ensuring the best decision (exercising or holding the option). Note that *intrinsic = np.maximum(K - np.exp(mvec), 0)*, which is the payoff in each node.

S	sigma	T	FD American	BS European	Early exercise value
36	0.2	1	4.4865	3.8443	0.6422
36	0.2	2	4.8483	3.7630	1.0853
36	0.4	1	7.1088	6.7114	0.3974
36	0.4	2	8.5140	7.7000	0.8140
38	0.2	1	3.2572	2.8519	0.4053
38	0.2	2	3.7515	2.9906	0.7609
38	0.4	1	6.1544	5.8343	0.3201
38	0.4	2	7.6748	6.9788	0.6960
40	0.2	1	2.3187	2.0664	0.2523
40	0.2	2	2.8893	2.3559	0.5335
40	0.4	1	5.3179	5.0596	0.2582
40	0.4	2	6.9232	6.3260	0.5972
42	0.2	1	1.6208	1.4645	0.1563
42	0.2	2	2.2165	1.8414	0.3752
42	0.4	1	4.5879	4.3787	0.2092
42	0.4	2	6.2501	5.7356	0.5145
44	0.2	1	1.1126	1.0169	0.0957
44	0.2	2	1.6931	1.4292	0.2639
44	0.4	1	3.9525	3.7828	0.1697
44	0.4	2	5.6466	5.2020	0.4446

Figure 16: Finite Difference for American options, BS for European options and Early Exercise Value.

Correctness test — FD American vs LSM American:

To demonstrate that the American finite difference implementation is correct, we compare its prices against the LSMC (Least Square Monte Carlo) estimates from Part 1(ii). If both methods are correctly implemented, they should converge to the same value.

S	sigma	T	FD American	LSM American	|Difference|
36	0.2000	1	4.4865	4.4710	0.0155
36	0.2000	2	4.8483	4.8321	0.0162
36	0.4000	1	7.1088	7.0988	0.0100
36	0.4000	2	8.5140	8.5084	0.0056
38	0.2000	1	3.2572	3.2507	0.0065
38	0.2000	2	3.7515	3.7415	0.0100
38	0.4000	1	6.1544	6.1616	0.0072
38	0.4000	2	7.6748	7.6692	0.0056
40	0.2000	1	2.3187	2.3138	0.0049
40	0.2000	2	2.8893	2.8848	0.0045
40	0.4000	1	5.3179	5.3263	0.0084
40	0.4000	2	6.9232	6.9133	0.0099
42	0.2000	1	1.6208	1.6121	0.0087
42	0.2000	2	2.2165	2.2057	0.0108
42	0.4000	1	4.5879	4.5762	0.0117
42	0.4000	2	6.2501	6.2477	0.0024
44	0.2000	1	1.1126	1.1194	0.0068
44	0.2000	2	1.6931	1.7016	0.0085
44	0.4000	1	3.9525	3.9539	0.0014
44	0.4000	2	5.6466	5.6403	0.0063

Figure 17: Testing.

All 20 absolute differences are below 0.05, confirming that the FD American implementation agrees with the LSM estimates.

Part (iii) — Full Replication of Table 1 (columns outside red box)

2.0/2.0

The columns outside the red box of Figure 2 are: Finite Difference American price, Closed-form European price, FD early exercise value (American − European), and the difference in early exercise value between the FD and LSMC methods.

S	sigma	T	FD American	Closed form European	FD Early exercise value	Difference in EE value
36	0.2000	1	4.4865	3.8443	0.6422	0.0155
36	0.2000	2	4.8483	3.7630	1.0853	0.0162
36	0.4000	1	7.1088	6.7114	0.3974	0.0100
36	0.4000	2	8.5140	7.7000	0.8140	0.0057
38	0.2000	1	3.2572	2.8519	0.4053	0.0065
38	0.2000	2	3.7515	2.9906	0.7609	0.0099
38	0.4000	1	6.1544	5.8343	0.3201	−0.0072
38	0.4000	2	7.6748	6.9788	0.6960	0.0056
40	0.2000	1	2.3187	2.0664	0.2523	0.0049
40	0.2000	2	2.8893	2.3559	0.5335	0.0045
40	0.4000	1	5.3179	5.0596	0.2582	−0.0085
40	0.4000	2	6.9232	6.3260	0.5972	0.0099
42	0.2000	1	1.6208	1.4645	0.1563	0.0087
42	0.2000	2	2.2165	1.8414	0.3752	0.0109
42	0.4000	1	4.5879	4.3787	0.2092	0.0118
42	0.4000	2	6.2501	5.7356	0.5145	0.0024
44	0.2000	1	1.1126	1.0169	0.0957	−0.0068
44	0.2000	2	1.6931	1.4292	0.2639	−0.0085
44	0.4000	1	3.9525	3.7828	0.1697	−0.0014
44	0.4000	2	5.6466	5.2020	0.4446	0.0063

Figure 18: Finite Difference American price, Closed-form European price, FD early exercise value (American − European), and the difference in early exercise value between the FD and LSMC methods.

The differences in early exercise value are small and alternate in sign, consistent with Table 1 of Longstaff and Schwartz (2001), confirming that both methods produce equivalent results.